

DEPT: Decoupled Embeddings for Pretraining Language Models

Alex Iacob; Lorenzo Sani; Meghdad Kurmanji; William F. Shen; Xinchu Qiu; Dongqi Cai;
Yan Gao; Nicholas D. Lane

01

Title & Authors

- DEPT: Decoupled Embeddings for Pretraining Language Models
- Alex Iacob; Lorenzo Sani; Meghdad Kurmanji; William F. Shen; Xinchu Qiu; Dongqi Cai; Yan Gao; Nicholas D. Lane
- They present a communication-efficient framework for heterogeneous pretraining.

02 DEPT in one slide (claims tied to Tables/Figs)

- Decouple embeddings from the transformer body to support training across heterogeneous data without a shared vocabulary
- Mitigate negative interference and the curse of multilinguality through isolated per-source updates
- Reduce communication proportional to fewer embedding updates and smaller parameter exchanges
- Deliver up to 20 percent average perplexity gains and improved downstream task scores
- Enable optimized per-source vocabularies that shorten sequences and strengthen the shared body
- They provide the first vocabulary-agnostic federated pretraining at billion scale with orders-of-magnitude lower communication

03

Motivation: Heterogeneous pretraining is hard

- Larger heterogeneous corpora improve generalization while introducing instability and cross-source interference
- The curse of multilinguality causes diminishing returns for low-resource languages
- Vocabulary dilution inflates sequences and wastes capacity under mixed domains
- Costly sampling tuning and filtering fail to resolve structural tokenizer mismatch
- Embedding contention concentrates updates on frequent sources and weakens rare domains

04

Motivation: Vocabulary dilution and embedding bottlenecks

- Tokenizer mismatch leads to over-fragmentation and longer sequences under UNIGRAM cross-entropy
- A single global vocabulary allows frequent sources to dominate while rare domains remain under-trained
- Embeddings and tied softmax dominate parameters and communication during distributed training
- Decoupled and specialized per-source embeddings isolate lexical statistics and reduce contention
- Optimized per-source vocabularies reduce sequence length and improve fit without global tuning

05

Embeddings as the bottleneck: evidence

- High tokenizer mismatch correlates with sequence inflation and excessive subword splits
- Shared vocabularies diffuse gradients and prioritize frequent sources over low-resource ones
- Embedding and output matrices dominate both parameter count and communication volume
- Tokenization mismatch compounds compute and bandwidth by increasing effective sequence length
- Decoupling targets the primary bottleneck by separating lexical and abstract pattern learning

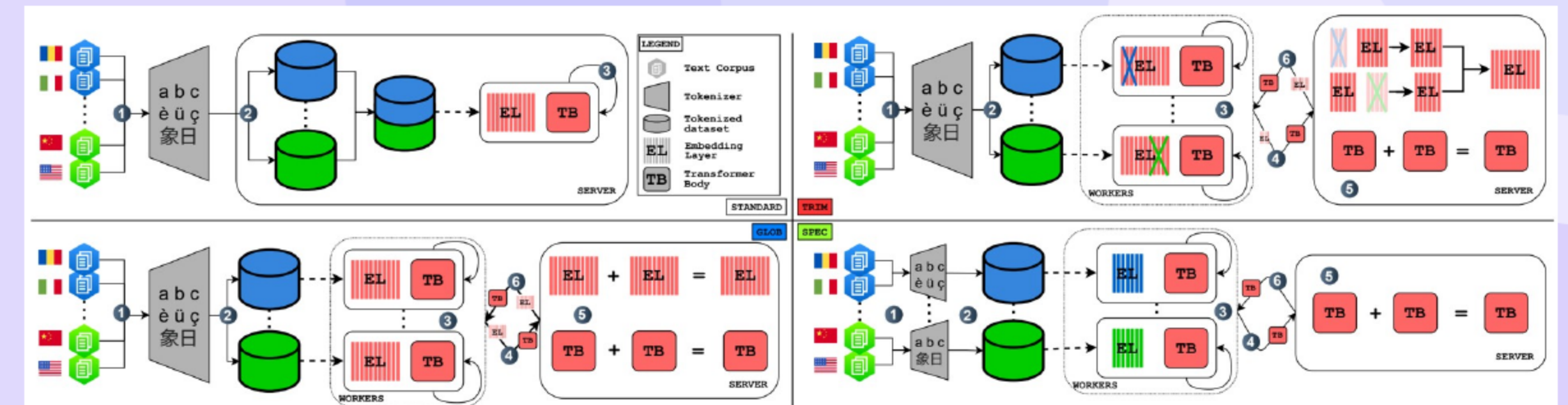
06

Why decoupling fixes interference

- Tokenizer mismatch inflates sequences and scatters updates across irrelevant embedding rows
- A shared vocabulary forces the body to model spurious subword boundaries instead of abstractions
- Per-source embeddings isolate updates so the shared body learns transferable patterns
- Gains concentrate where tokenizer mismatch is highest under unigram cross-entropy
- Shorter sequences and focused gradients stabilize optimization under heterogeneity

07

DEPT overview: variants + compact setup



- GLOB shares a global vocabulary and embeddings while TRIM selects source-relevant rows and SPEC syncs only the shared body
- Clients train locally with InnerOPT and aggregate with a FedAvg-style OuterOPT on the server
- Sources operate as silos and exchange only selected parameters to reduce memory and communication

08

Baselines and citations

- STD and Local SGD — Lin, 2020; Stich, 2019
- FedAvg — McMahan, 2017
- ACT (active forgetting) — Chen, 2023
- Tokenizers/corpora: mT5 SentencePiece; The Pile — Xue, 2021; Black, 2022
- Communication/memory context: DiLoCo; ZeRO — Douillard, 2023; Rajbhandari, 2020

09

Result highlights (what to remember)

- Lower perplexity and stronger downstream performance across diverse sources
- Reduced memory and communication through fewer and smaller parameter exchanges
- Improved stability with constrained activation and model norms under identical local settings
- Consistent benefits at higher tokenizer mismatch and in out-of-distribution evaluations

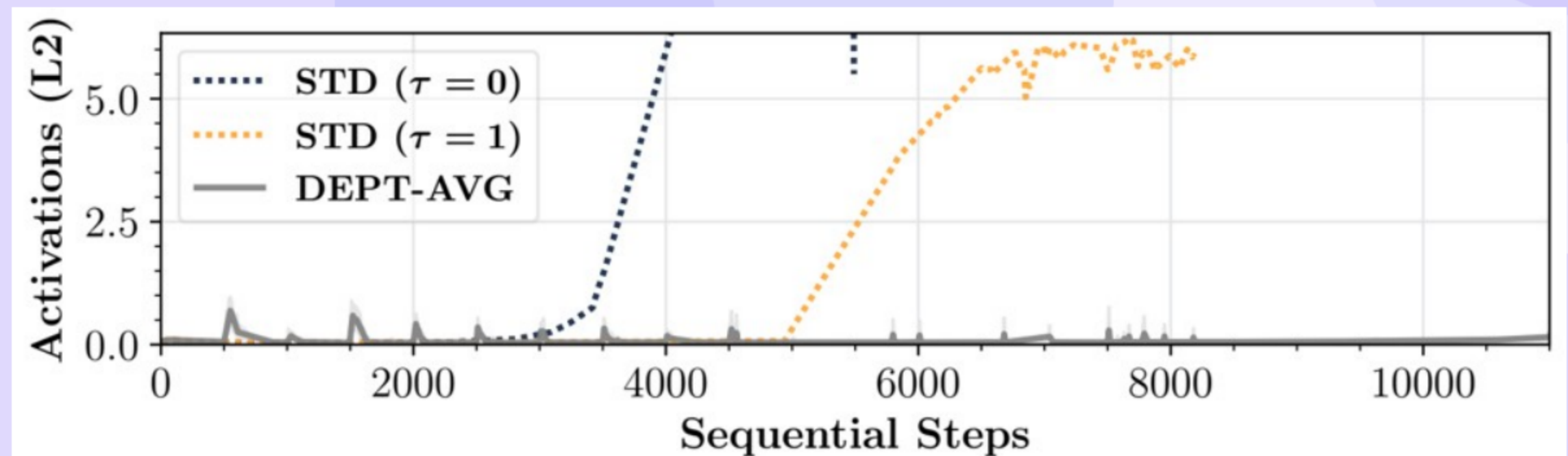
10

Most surprising finding

- SPEC and TRIM strengthen the shared body without sharing a vocabulary or synchronizing embeddings
- Average perplexity improves by 10–20% while communication drops by 500–700×.
- Embedding memory shrinks 4–5× at billion-parameter scale.
- Removing embedding communication still yields a more adaptable and transferable body

11

RQ1: Robust to heterogeneity (Fig. 3)



- DEPT maintains low activation and model norms under identical local hyperparameters
- Baselines require learning rate reductions to avoid divergence during training
- Outer optimization noise provides regularization that curbs norm inflation
- Stability persists across multilingual and multi-domain configurations

12 RQ2: Communication and memory efficiency (Table 2; Fig. 5)

Type	#Blocks	Method	N_{local}	T	$ \mathcal{V}_k \pm \sigma$	$ \mathcal{V}_k \times d_{model}$	$\overline{\mathcal{M}}_k (\downarrow)$	Per-step Comms Cost ($\downarrow$)
Multilingual	12	STD	5×10^3	1	250 112	192M	278M (1 $\times$)	278M (1 $\times$)
Multilingual	12	GLOB	500	10	250 112	192M	278M (1 $\times$)	0.56M (0.002 $\times$)
Multilingual	12	TRIM	500	10	216 135 $\pm$ 27 160	166M	252M (0.92 $\times$)	0.5M (0.002 $\times$)
Multilingual	12	SPEC	500	10	216 135 $\pm$ 27 160	166M	252M (0.92 $\times$)	0.17M (0.0006 $\times$)
Multilingual	12	SPEC-OPT	500	10	50 257 $\pm$ 0	38.6M	125M (0.45 $\times$)	0.17M (0.0006 $\times$)
Multilingual-B	24	STD	7×10^3	1	250 112	512.2M	1.71B (1 $\times$)	1.71B (1 $\times$)
Multilingual-B	24	SPEC-OPT	500	14	50 257 $\pm$ 0	102.9M	1.3B (0.76 $\times$)	2.4M (0.001 $\times$)
Multi-domain	12	STD	5×10^3	1	50 257	38.6M	125M (1 $\times$)	125M (1 $\times$)
Multi-domain	12	GLOB	500	10	50 257	38.6M	125M (1 $\times$)	0.25M (0.002 $\times$)
Multi-domain	12	TRIM	500	10	45 554 $\pm$ 9462	35M	121M (0.97 $\times$)	0.24M (0.002 $\times$)
Multi-domain	12	SPEC	500	10	45 554 $\pm$ 9462	35M	121M (0.97 $\times$)	0.17M (0.001 $\times$)
Multi-domain	24	STD	13.5×10^3	1	50 257	51.4M	350M (1 $\times$)	350M (1 $\times$)
Multi-domain	24	GLOB	500	27	50 257	51.4M	350M (1 $\times$)	0.7M (0.002 $\times$)
Multi-domain	24	TRIM	500	27	45 554 $\pm$ 9462	46.6M	345.2M (0.97 $\times$)	0.69M (0.002 $\times$)
Multi-domain	24	SPEC	500	27	45 554 $\pm$ 9462	46.6M	345.2M (0.97 $\times$)	0.6M (0.002 $\times$)

- All DEPT variants reduce per-step communication via more local steps and smaller vocabularies; SPEC removes embedding communication entirely.
- TRIM cuts memory by discarding irrelevant rows while preserving convergence
- Communication and memory improvements scale to billion-parameter models in cross-silo settings. Aggregation via FedAvg (McMahan, 2017); comms context vs. ZeRO/DiLoCo (Rajbhandari, 2020; Douillard, 2023).

14 Downstream gains and plasticity (Table 7; Fig. 4)

- Downstream accuracy improves by several percentage points with TRIM strongest overall
- Adaptation is faster from random initialization and to new or low-resource languages
- Restricted vocabularies can enhance shift robustness without discarding updates
- Perplexity drops more quickly under distribution shift with stable convergence

13 RQ3: Zero-shot generalization—stronger bodies (Tables 3–4, 10, 12)

Name (UNIGRAM-CE)	DM (6.9)	EN (7.9)	EP (10)	FL (7.8)	GH (7.9)	CC (7.9)	PA (8.2)	SE (7.7)	PP (9.1)	WK (8.2)	AX (7.7)	UB (7.8)	PC (8)	NH (8.1)	GU (7.7)	HN (7.7)	UI-OOD (10)	AVG (8.1)
STD ($\tau = 0$)	5.5	44.8	93.5	30.9	8.1	79.6	46.6	23.4	126.6	58.2	14.3	34.1	22.3	58.9	76.3	65.2	163.6	56
STD ($\tau = 1$)	5	30.6	49.5	20.6	6	56.2	30.9	16.8	81.2	39.1	11	23.7	16.1	39.3	54.6	46.9	99	36.9
ACT	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
GLOB	4.8	25.7	38.2	17.3	5.4	47.7	25.7	14.7	68.3	32.7	9.9	20	14	32.2	46.5	39.8	94.8	31.6
TRIM	4.8	27.3	39.5	18.5	5.6	51.2	27.8	15.4	71.8	35.1	10.3	21.7	14.8	35.1	49.1	42.2	95.7	33.3
SPEC	4.8	26.7	36.8	18.2	5.5	50.1	27.1	15.1	69.1	34.2	10.1	21.1	14.5	34.3	48.5	41.7	97.6	32.7
SPEC-OPT	4.7	25.9	35	17.5	5.4	48.3	26.1	14.7	66.6	32.8	9.9	20.4	14.1	32.9	47.3	40.5	88.6	31.2
Min Imp (%)	3.7	10.6	20.2	10.1	7.4	8.9	10.3	8.4	11.5	10.3	7	8.6	8.2	10.6	9.9	10	1.4	9.7
Max Imp (%)	4.2	15.7	29.3	16.3	11	15.1	16.9	12.9	17.9	16.5	10.6	15.7	13.3	18	14.7	15.2	10.5	15.3

- DEPT wins the large majority of comparisons with the largest gains at higher tokenizer mismatch
- The Pile shows sizable average validation improvements with strongest results under optimized vocabularies
- MC4 yields double-digit gains in distribution and out of distribution across languages
- TRIM matches or exceeds global baselines despite fewer parameters and lower communication

15 Related work (concise): scaling, alignment, and vocab design

- Scaling multilingual language models broadens coverage while causing uneven performance under imbalance
- Parallel supervision and transfer improve low-resource learning with risks of instability and forgetting
- Vocabulary design with temperature control and language clustering improves coverage with added complexity
- Active forgetting reinitializes embeddings to boost plasticity during adaptation
- Federated aggregation strategies enable cross-silo training under heterogeneous data and tokenizers

- SPEC requires a final global embedding matrix for practical deployment
- Zero-shot embedding transfer and vocabulary matching offer viable stitching strategies
- Continued pretraining can equip decoupled models with a shared output space
- Privacy-preserving tokenizers remain an open opportunity for siloed sources
- Scaling laws under heterogeneous tokenization require further empirical study

- They show decoupling embeddings enables efficient, robust heterogeneous pretraining at scale.
- They report several-fold embedding memory drops with large communication reductions.
- They find improved generalization and plasticity across domains and languages without shared vocabularies.
- They demonstrate vocabulary-agnostic federated pretraining at billion-parameter scale.
- They outline practical deployment via continued pretraining and embedding stitching.

Thank You

Questions & Discussion